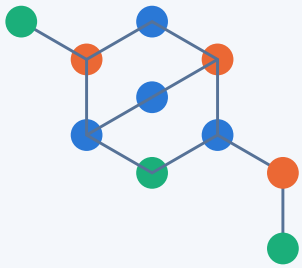
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ChemFlow

An Auditable Workflow for Docking-Based Virtual Screening with Redocking Validation and Chemometric Analysis



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ABSTRACT

Docking-based virtual screening combines ligand and receptor preparation, docking, pose validation and interaction analysis, steps that are usually run by hand in separate programs and are difficult to audit afterwards. ChemFlow is an open-source (MIT) Python workflow that integrates these steps behind one command line, one web interface and one library. It prepares ligands and receptors, runs AutoDock Vina with seeded replicates, validates the setup by redocking the crystallographic ligand and measuring in-place heavy-atom RMSD, scores interaction similarity to that native ligand, summarises physicochemical and ADMET properties, performs chemometric analysis (PCA and hierarchical clustering), and offers an optional GC–MS module. Version 3.1.0 implements each derived quantity as a documented formula and cross-checks the redocking RMSD with an independent implementation. Redocking of the native ligands of three public complexes (stored Vina run) yields minimum in-place RMSDs of 0.62, 0.85 and 1.74 Å, and BIOVIA Discovery Studio reproduces the in-place RMSD to within 4.7×10^{-5} Å. ChemFlow introduces no new scoring function and has not been evaluated in a prospective screen; the similarity, PCA and GC–MS examples use constructed or simulated data.

Keywords chemometrics; GC–MS; interaction similarity; molecular docking; redocking validation

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ChemFlow 3.1.0, MIT licence

STATUS
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CODE
`github.com/Arifmaulanaazis/chemflow`

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1 Introduction

Structure-based virtual screening estimates which small molecules in a library are likely to bind a protein of known structure. In practice the calculation is a pipeline. Ligand structures must be converted from text (SMILES) into three-dimensional, protonated, charge-assigned files; the receptor must be cleaned and typed; a search box must be placed; a docking engine such as AutoDock Vina [1, 2] must be run, ideally with replicates; and the setup should be checked by redocking the co-crystallised ligand before any prediction is trusted [3]. Afterwards the poses are interpreted by comparing their contacts with those of the native ligand and by locating the compounds in descriptor space.

Each of these steps has good tools, and they are often combined through graphical front ends [4, 5]. Records of the choices made are harder to obtain. When the steps are executed by hand, the exact random seed, the force-field fallback that was used, the box size, the atom-typing rules and the version of the docking binary are rarely stored together with the numbers that came out. A further difficulty is that several standard quantities contain a choice that changes the result. Root-mean-square deviation (RMSD) can be computed after superposing two structures or in place; the former is a measure of shape and the latter of placement. A similarity score can be symmetric or directional. A standard deviation can use n or $n - 1$ in the denominator.

ChemFlow implements the pipeline as one auditable program. Its design goals are:

1. **One workflow, three interfaces.** A command-line tool, a local web interface and a Python library drive the same code, so a run started in one can be resumed in another.
2. **Explicit, documented formulas.** Every derived quantity is a short function with a documented definition (Section 4).
3. **Validation built in.** Redocking of the native ligand, with in-place RMSD and an independent cross-check, is a stage of the pipeline.
4. **Provenance.** Seeds, parameters, checkpoints and per-pose records are written next to the results, and an interrupted run can be resumed.

Scope. ChemFlow 3.1.0 introduces no new scoring function, force field or statistical estimator: the scientific methods are those cited, and its contribution is their integration, parameterisation and auditing. The software has not been evaluated in a prospective screen, and no statement about the biological activity of any compound is made.

2 Related work and positioning

Docking engines and front ends. AutoDock Vina [1] replaced the earlier AutoDock4 scoring and search [6] with an empirical scoring function and an iterated local search, and version 1.2 added further scoring options and tooling [2]. AutoDockTools and the associated protocols [5] and the PyRx environment [4] wrap docking in graphical or scripted workflows. ChemFlow does not replace these: it calls an unmodified Vina executable and adopts the AutoDock4 atom-type vocabulary [6] for receptor files.

Chemistry toolkits. RDKit [7] provides molecule handling, conformer generation (ETKDG [8]), force fields (MMFF94 [9], UFF [10]), Gasteiger charges [11] and descriptors; Open Babel [12] converts file formats. Public databases supply structures [13] and compound records [14].

ADMET. ADMETlab 3.0 [15] is a web platform for ADMET prediction. ChemFlow retrieves or imports its output and applies a rule table; it does not implement an ADMET model.

Positioning. ChemFlow provides workflow integration and an audit trail: spreadsheet-driven batch preparation, seeded replicates, a mandatory validation stage, per-pose provenance, resume support, an interaction-similarity stage, and chemometric summaries, all under one versioned and tested codebase.

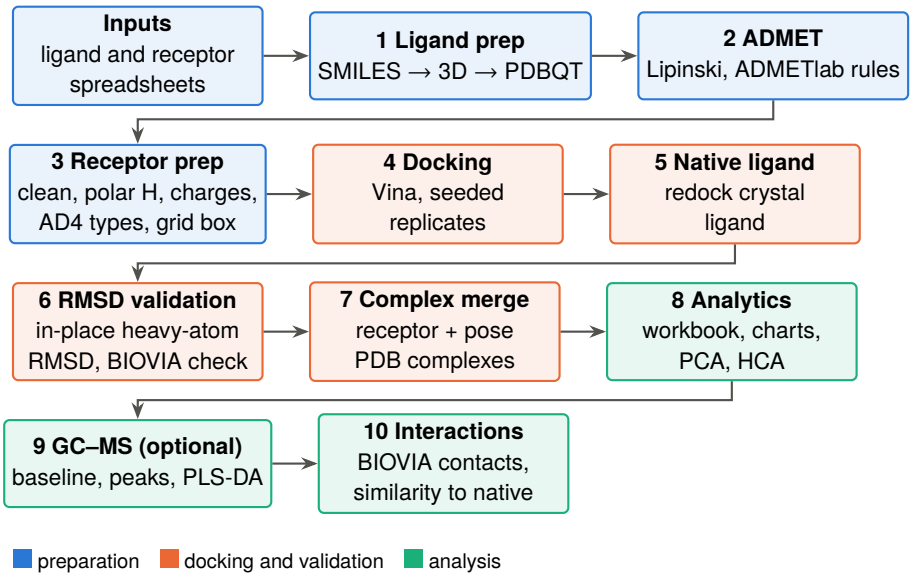


Figure 1. The ChemFlow pipeline (stages follow `Pipeline.run` in `chemflow/pipeline.py`). Colour marks the role of a stage (blue: preparation; orange: docking and validation; green: analysis), and every box carries its own label. Stage 9 is optional and runs independently of docking; stage 10 consumes interaction tables that the user exports from BIOVIA Discovery Studio after docking.

3 System overview

3.1 Pipeline

Figure 1 shows the ten stages executed by `chemflow/pipeline.py`. Two spreadsheets define a run: one lists ligands (SMILES, or compound names resolved through PubChem [14], with an optional group label), and one lists receptors (PDB code, search-box centre and size, optionally several sites per structure). Stages write checkpoints under a `_state/` directory, so an interrupted run can be resumed. A `--rmsd-only` mode skips test ligands, ADMET and analytics and performs only the validation.

3.2 Interfaces and architecture

Table 1 lists the main components. The web interface is a FastAPI service with a React/TypeScript front end that uses the NGL viewer [16] for structures. It serves one local user: sessions use an HTTP-only cookie, state-changing requests carry a CSRF token compared in constant time, jobs are queued in SQLite and executed as child processes of `python -m chemflow` without a shell, and progress is streamed to the browser as server-sent events. The library exposes the same pipeline for scripting. The `chemflow/` package of version 3.1.0 contains 91 Python modules (17 617 lines), and `tests/` contains 61 test files.

4 Methods

4.1 Ligand preparation

Each SMILES is parsed with RDKit [7]. The largest fragment is kept and the structure is neutralised (`LargestFragmentChooser` and `Uncharger`), explicit hydrogens are added, and a 3D conformer is embedded with ETKDG [8], trying version 3 first and version 2 as a fallback, with a fixed random seed of 42. The conformer is minimised with MMFF94 [9] for up to 1000 iterations, falling back to UFF [10] when MMFF94 parameters are unavailable. The record states which force field was actually used. The structure is then converted to PDBQT by Open Babel [12] with Gasteiger partial charges [11]. Only one conformer per ligand is generated, and protonation states, tautomers and stereoisomers are not enumerated.

Table 1. Main components of ChemFlow 3.1.0 and what each is responsible for.

Component (under `chemflow/`)	Responsibility
`chem/ligand_preparer.py`	Salt stripping, neutralisation, 3D embedding, force-field minimisation, charges, PDBQT
`chem/receptor_preparer.py`, `kollman_charges.py`, `atom_typing.py`	Receptor cleaning, polar hydrogens, Kollman-style charge table, AutoDock4 atom types
`chem/descriptors.py`, `admet/`	Lipinski descriptors; ADMETlab retrieval or import, deduplication, rule flags
`docking/grid_box.py`	Search-box suggestion from the native ligand
`docking/vina_manager.py`, `vina_runner.py`, `docking_matrix.py`	Vina download and invocation, ligand × receptor × replicate matrix, checkpoints, replicate statistics
`docking/rmsd_validation.py`, `rmsd_pair.py`, `rmsd_backend.py`	Native-ligand RMSD, atom mapping, backend selection (RDKit or BIOVIA)
`docking/merge.py`	Receptor–pose complexes with JSON sidecars
`interaction/`, `similarity/`	BIOVIA automation and interaction-table reader; similarity to the native ligand
`analytics/`	PCA, hierarchical clustering, heat maps, group statistics, charts
`gcms/`	Readers, preprocessing, statistics, library matching for GC–MS
`web/`, `frontend/`	FastAPI service, job queue, React interface

Gasteiger charges. The Gasteiger–Marsili procedure [11] assigns partial charges by iteratively equalising orbital electronegativity. Each atom i has an electronegativity that depends on its own charge q_i,

$$\chi_i(q_i) = a_i + b_i\, q_i + c_i\, q_i^2, \tag{1}$$

where a_i, b_i, c_i are tabulated for the atom's hybridisation state. In iteration k, a bonded pair (i, j) exchanges charge

$$\Delta q_{i\leftarrow j}^{(k)} = \frac{\chi_j - \chi_i}{\chi_i^+}\,\alpha^k, \qquad \alpha = \tfrac{1}{2}, \tag{2}$$

where $\chi_i^+ = \chi_i(+1)$ is the electronegativity of the cation, which normalises the transfer, and the damping factor α^k makes the process converge in a few iterations. Charge moves from the less to the more electronegative atom. In ChemFlow the charge column of the ligand PDBQT is written by Open Babel; the RDKit Gasteiger charges computed during preparation are not copied into it. AutoDock Vina's scoring function does not use partial charges in its energy terms [1], so the charge column matters mainly for file compatibility and downstream use.

Lipinski descriptors. Molecular weight, Crippen logP, and hydrogen-bond donor and acceptor counts are computed with RDKit. The number of rule-of-five violations [17] is

$$V = [\,M_w > 500\,] + [\,\log P > 5\,] + [\,\mathrm{HBD} > 5\,] + [\,\mathrm{HBA} > 10\,], \tag{3}$$

where $[\,\cdot\,]$ is 1 if the condition holds and 0 otherwise. A ligand passes when $V \le 1$.

4.2 Receptor preparation and atom typing

Two receptor files are produced for each structure, because docking and complex building need different things. The *docking* receptor has waters and heteroatoms removed, only polar hydrogens kept, Kollman-style partial charges assigned from a lookup table, AutoDock4 atom types [6] written, and torsion tags removed so that Vina treats it as rigid. The *merge* receptor is a clean PDB without added hydrogens or charges, used later to build complexes.

Table 2. AutoDock4 atom-type rules applied by `chem/atom_typing.py`.

Element	Condition	Type
H	bonded to a non-carbon atom (polar) / otherwise	HD / H
C	aromatic / otherwise	A / C
N	aromatic; or non-aromatic with no carbonyl neighbour / non-aromatic next to a carbonyl (amide-like)	NA / N
O	always (acceptor)	OA
S	double bond to oxygen / otherwise	S / SA

Charges. The table (`chem/kollman_charges.py`) follows the AMBER convention [18, 19]: backbone atoms use fixed values (for example N −0.4157, H +0.2719, Cα +0.0337, C +0.5973, O −0.5679), side chains use a curated table for the 20 standard residues, histidine tautomers (HID, HIE, HIP) and CYX are recognised from the hydrogens present, and the N- and C-terminus are corrected. Atoms absent from the table receive 0.0 by default or RDKit Gasteiger charges on request. This is a representative table, not a full implementation of a published force field, and many non-polar carbons are 0.0 because non-polar hydrogens are merged. This limitation is summarised in Section 7.

Atom types. Table 2 lists the local rules that assign AutoDock4 types. They are deliberately simple and use RDKit's aromaticity and neighbour information.

4.3 Search box

The box is derived from the extent of the native ligand. With E the longest coordinate range of the native ligand, padding $p = 8$ Å, minimum side $s_{\min} = 18$ Å and cap $s_{\max} = 30$ Å,

$$s = \min\big(\lceil \max(E + p,\ s_{\min}) \rceil,\ s_{\max}\big). \tag{4}$$

The padding leaves room for the ligand to move; the minimum avoids a box too small to sample; and the cap limits cost and the loss of search efficiency in very large boxes. For example $E = 5, 12, 24, 40$ Å give $s = 18, 20, 30, 30$ Å. The native ligand is the one whose centre is closest to the user-supplied box centre, within 8 Å; if no native ligand is found, a fallback side of 20 Å is used.

4.4 Docking and replicate statistics

Docking runs the unmodified AutoDock Vina binary [1, 2]; the version is selectable and the binary is downloaded from the project's release page into a cache. For each ligand–receptor pair, R replicates are run with seeds $\text{seed}_r = \text{seed}_0 + r - 1$, so every result can be repeated. Let a_r be the affinity (kcal mol^{-1}) of the top-ranked mode in replicate r. ChemFlow reports

$$\bar{a} = \frac{1}{R}\sum_{r=1}^{R} a_r, \qquad s_a = \sqrt{\frac{1}{R}\sum_{r=1}^{R}(a_r - \bar{a})^2}, \qquad a_{\text{best}} = \min_r a_r, \tag{5}$$

and, for a native ligand with affinity ΔG_{nat}, the difference $\Delta_{\text{nat}} = a_{\text{best}} - \Delta G_{\text{nat}}$ (negative means stronger than the native ligand). Two conventions apply. First, the standard deviation uses R in the denominator (population form), not $R - 1$: for $a = (-7, -8, -9)$ this gives 0.816 whereas the sample form gives 1.000. Second, $s_a = 0$ when only one replicate exists, which is an absence of information rather than evidence of stability.

4.5 Redocking validation by in-place RMSD

The native ligand is extracted from the crystal structure, redocked into the same box, and each docked pose is compared with the crystal pose. For N heavy atoms with docked coordinates $\mathbf{x}_i$ and crystal coordinates

$\mathbf{y}_i$,

$$\mathrm{RMSD} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{x}_i - \mathbf{y}_{\pi(i)}\right\|^2}, \tag{6}$$

where π is the atom correspondence between the two molecules. Two choices define this quantity. *No superposition*: the poses are not rotated or translated before comparison. This is appropriate for docking validation because docked and crystal ligands live in the same receptor frame, and an optimal superposition would hide placement error and reward only shape. RDKit's `CalcRMS` computes exactly this in-place value and takes molecular symmetry into account. *Atom mapping*: π is the symmetry-aware matching found by RDKit; bond orders of the crystal ligand are assigned from the RCSB chemical-component template [13]; and pairs with differing atom or bond counts are rejected. A greedy nearest-atom fallback exists for hard cases but is flagged as approximate.

Classification and pose selection. Following the usual criteria [3], a pose is classed

$$\text{class} = \begin{cases} \text{good}, & \mathrm{RMSD} < 2.0\,\text{Å}, \\ \text{acceptable}, & 2.0\,\text{Å} \le \mathrm{RMSD} < 3.0\,\text{Å}, \\ \text{poor}, & \mathrm{RMSD} \ge 3.0\,\text{Å}, \end{cases} \tag{7}$$

with strict inequality at 2.0 Å. All poses of all replicates are scored. The pose with minimum RMSD is the *selected best* and the receptor setup is judged on it, but the RMSD of the pose with the best docking affinity is always reported alongside. Taking the minimum over many poses makes a threshold easier to meet and is not evidence that the scoring function ranks the correct pose first.

Independent cross-check. When BIOVIA Discovery Studio is available, its heavy-atom RMSD routine (`CalculateHeavyAtomRmsd`) is evaluated on the same mapped pair; the BIOVIA value is accepted only if it agrees with RDKit to 0.002 Å on an unaligned pair, and otherwise the RDKit value is used with the reason recorded.

4.6 Interaction similarity to the native ligand

Contacts between ligand and protein are exported from BIOVIA Discovery Studio (the automation drives the GUI on Windows and locks the mouse and keyboard during the run; pressing Esc three times aborts). Intra-ligand and intra-protein rows are discarded. Let A_R be the set of residues, identified by residue number and name, that contact the reference (native) ligand, A_T the same for the test ligand, and I_R, I_T the sets of (residue, interaction-type) pairs. Following Pratama *et al.* [20],

$$S_{\mathrm{res}} = \frac{|A_T \cap A_R|}{|A_R|} \times 100\%, \tag{8}$$

$$S_{\mathrm{type}} = \frac{|I_T \cap I_R|}{|I_R|} \times 100\%, \tag{9}$$

$$S = \tfrac{1}{2}\, S_{\mathrm{res}} + \tfrac{1}{2}\, S_{\mathrm{type}}. \tag{10}$$

S_{res} asks whether the test ligand touches the same residues as the native ligand, regardless of how; S_{type} is stricter and also requires the same kind of interaction (hydrogen bond, hydrophobic, and so on); S averages the two with equal weights. The affinity difference plotted against S is

$$\Delta G = a_{\mathrm{test}} - a_{\mathrm{ref}}. \tag{11}$$

Because every denominator is a property of the *reference*, the score is directional: contacts that the test ligand makes beyond the native set are not penalised. Residue keys do not include the chain, so equal numbers and names on different chains would collide; a reference with no contacts raises an error rather than dividing by zero.

4.7 Descriptor and ADMET summaries

For ADMET, ChemFlow retrieves results from ADMETlab 3.0 [15] in batches (with back-off on HTTP 429) or imports a CSV or Excel file, deduplicates molecules by canonical SMILES, and applies a rule table that assigns each endpoint one of the flags GREEN, YELLOW, RED, NA or INFO. The table cites its literature sources endpoint by endpoint (for example oral-bioavailability descriptors [21]). A category score is the mean of the scored flags,

$$\text{score}_c = \frac{1}{n_c} \sum_{e \in c} f(e), \qquad f(\text{GREEN}) = 1,\ f(\text{YELLOW}) = \tfrac{1}{2},\ f(\text{RED}) = 0, \tag{12}$$

where the sum runs over endpoints e of category c that carry a scored flag; descriptive (INFO) and missing (NA) endpoints are excluded. The numbers $1, \frac{1}{2}, 0$ are an ordinal convenience, not probabilities.

4.8 Chemometrics for property space

Scaling and PCA. Let x_{ij} be feature j of sample i, with column mean μ_j and population standard deviation σ_j. The scaled value is

$$\tilde{x}_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j^{\,p}}, \qquad p = 1 \text{ (auto)},\ \tfrac{1}{2} \text{ (Pareto)},\ 0 \text{ (centre)}, \tag{13}$$

with $\sigma_j = 0$ replaced by 1. Auto-scaling gives every feature unit variance and is the default; Pareto scaling is a compromise that damps large features without erasing their scale; centring only removes the mean. Principal component analysis [22, 23] then finds orthogonal directions of maximal variance. If λ_k is the k-th eigenvalue of the covariance matrix of $\tilde{X}$, component k explains the fraction

$$\text{EVR}_k = \frac{\lambda_k}{\sum_m \lambda_m}. \tag{14}$$

The main property-space analysis uses four features (molecular weight, logP, HBD, HBA); two or three components are retained and reduced automatically when samples or features are too few. A 95% confidence ellipse can be drawn per group from the covariance matrix Σ of its two-dimensional scores: it is the set $\{\mathbf{z} : (\mathbf{z} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{z} - \boldsymbol{\mu}) \le \chi^2_{2,0.95}\}$ with $\chi^2_{2,0.95} = 5.991$, whose semi-axes are $\sqrt{5.991\,\lambda_i}$ along the eigenvectors of Σ. At least three points are required, and an ellipse from a handful of points is only a rough guide.

Hierarchical clustering. Features are z-scored and clustered with Ward's method [24] on Euclidean distance, as implemented in SciPy [25]. Ward's criterion merges, at each step, the two clusters A, B whose union increases within-cluster variance least:

$$\Delta(A, B) = \frac{|A|\,|B|}{|A| + |B|} \left\| \mathbf{c}_A - \mathbf{c}_B \right\|^2, \tag{15}$$

where $\mathbf{c}_A, \mathbf{c}_B$ are cluster centroids.

4.9 Optional GC–MS analysis

The GC–MS module accepts chromatograms in several formats (Shimadzu text/Excel, CSV, ANDI `.cdf`, mzML, mzXML) and processes total-ion traces in the steps below.

Baseline. The baseline $\mathbf{z}$ is the asymmetric least-squares smoother, a weighted variant of the Whittaker penalised smoother [26]:

$$\mathbf{z} = \arg\min_{\mathbf{z}} \sum_i w_i \,(y_i - z_i)^2 + \lambda \sum_i (\Delta^2 z_i)^2, \qquad w_i = \begin{cases} p, & y_i > z_i, \\ 1 - p, & y_i < z_i, \end{cases} \tag{16}$$

with $\lambda = 10^7$, $p = 0.001$ and 10 iterations. The first term keeps $\mathbf{z}$ close to the signal, the second penalises curvature (Δ^2 is the second difference), and the asymmetric weights make points above the current baseline (peaks) count very little, so that the baseline follows the valleys.

Smoothing, noise and peaks. The signal is smoothed with a Savitzky–Golay filter [27] (default window 0.02 min, cubic polynomial). Noise is estimated robustly from the residual r of a short filter as $\hat{\sigma} = 1.4826\,\mathrm{median}|r - \mathrm{median}(r)|$, where 1.4826 makes the median absolute deviation consistent with a normal standard deviation. Peaks are found with SciPy's `find_peaks` with minimum prominence $\max(5\hat{\sigma},\ 0.005\, y_{\max})$; areas are trapezoidal integrals at relative height 0.95. Retention times are aligned across samples by cross-correlating square-root-compressed signals against a medoid reference (maximum shift 0.3 min, two passes), peaks within 0.05 min are merged into one feature, and gaps are filled.

Normalisation and transformation. Each sample is first scaled to a total of 100. Probabilistic quotient normalisation (PQN) [28] then divides each sample i by the median quotient to a reference spectrum (the feature-wise median of the sample set),

$$\tilde{x}_{ij} = \frac{x_{ij}}{\mathrm{median}_j\left(x_{ij}/x_j^{\mathrm{ref}}\right)}, \tag{17}$$

which makes the correction robust to a few large changes. The default is total-sum normalisation; $\log_{10}$ (with an offset of half the smallest positive value), square-root and no transformation are available.

Univariate tests. For two classes the module reports Welch's t-test [29], the Mann–Whitney test [30], $\log_2$ fold change and Cohen's d; for three or more classes, ANOVA and Kruskal–Wallis; and always η^2 and the Benjamini–Hochberg [31] adjusted q-values. With m tests and ascending $p_{(1)} \le \cdots \le p_{(m)}$,

$$q_{(k)} = \min\Big(1,\ \min_{l \ge k} \frac{m}{l}\, p_{(l)}\Big). \tag{18}$$

A feature with $q < 0.05$ is expected to carry a false-discovery proportion of at most 5% among the features called significant.

PLS-DA, VIP and permutation test. Supervised discrimination uses partial least squares [32, 33] regressing a one-hot class matrix, with the number of components chosen by cross-validated $Q^2 = 1 - \mathrm{PRESS}/\mathrm{SS}_{\mathrm{tot}}$ (prediction error sum of squares over total sum of squares; values near 1 mean good prediction). Variable importance in projection is

$$\mathrm{VIP}_j = \sqrt{p\,\frac{\sum_a \mathrm{SS}_a \left(w_{ja}/\|\mathbf{w}_a\|\right)^2}{\sum_a \mathrm{SS}_a}}, \qquad \mathrm{SS}_a = q_a^2\, \mathbf{t}_a^{\top}\mathbf{t}_a, \tag{19}$$

where p is the number of features, $\mathbf{w}_a$ the weights, $\mathbf{t}_a$ the scores and q_a the Y-loading of component a; the mean of VIP_j^2 is 1, so VIP > 1 marks above-average features. A permutation test shuffles the labels n times and reports

$$p_{\mathrm{perm}} = \frac{n_{\ge} + 1}{n + 1}, \tag{20}$$

where $n_{\ge}$ counts shuffles that score at least as well as the real labelling; the +1 terms prevent a reported p of zero. Linear discriminant analysis on principal-component scores is also provided. Models are blocked when a class has fewer than two samples or fewer than six samples are available.

Table 3. Redocking of native ligands (27 stored poses; in-place heavy-atom RMSD, Å). "Min" is the selected-best pose; "mode 1" is the pose with the best Vina affinity. Class counts use Eq. (7) over the nine modes.

PDB	Ligand	Heavy atoms	Min RMSD (Å)	Min mode	Mode-1 RMSD (Å)	Good / acc. / poor	Class of mode 1
1UWH	BAX	32	0.618	1	0.618	2 / 2 / 5	good
4EY7	E20	28	0.851	1	0.851	1 / 3 / 5	good
1HCK	ATP	31	1.738	2	2.742	2 / 2 / 5	acceptable

Outlier screening and retention index. Hotelling's T^2 [34] of sample i in the space of a retained components is $T_i^2 = \sum_k t_{ik}^2 / s_k^2$ (scores t, score variance s_k^2). For training observations, an upper $(1-\alpha)$ limit is $\frac{(n-1)^2}{n} B_{1-\alpha}\left(\frac{a}{2}, \frac{n-a-1}{2}\right)$, where B is a beta quantile [35]. Temperature-programmed Kováts retention indices use linear interpolation between n-alkanes [36]:

$$\mathrm{RI} = 100 \left[n + \frac{t_x - t_n}{t_{n+1} - t_n} \right], \tag{21}$$

with t_x the retention time of the analyte and t_n, t_{n+1} those of the alkanes with n and $n+1$ carbons that bracket it. Compound names are assigned only from a user-supplied library by retention time or index; the module does not perform spectral library matching.

4.10 Software environment

Results in Section 5 were produced with Python 3.14.6, RDKit 2026.03.5 [7], NumPy 2.5.2 [37], pandas 3.0.5, SciPy 1.18.1 [25], scikit-learn 1.9.1 [23] and Matplotlib [38] on Windows 10 (10.0.19045). The archived docking run used AutoDock Vina 1.2.7 with seed 42.

5 Results

5.1 Software verification

The pytest suite of the repository comprises 950 tests: 949 passed, 1 was skipped (a test that needs symbolic-link privileges on Windows) and none failed, in 95 s. The front-end suite (Vitest) reports 18 of 18 tests passing in 4 files. Line coverage has not been measured, and many tests use synthetic fixtures and mock interfaces, so passing tests show consistency with the intended behaviour rather than scientific correctness.

5.2 Redocking validation on three public complexes

The validation stage was exercised on a stored Vina 1.2.7 run (seed 42, one replicate, nine modes) for the native ligands of three public PDB entries. The 27 docked/crystal pairs (3 complexes × 9 modes) were re-evaluated without re-docking. The in-place RMSD of Eq. (6) was recomputed independently with NumPy from the stored, atom-ordered coordinates and agrees with the stored RDKit values to machine precision; BIOVIA Discovery Studio's `CalculateHeavyAtomRmsd` agrees with them to at most 4.7×10^{-5} Å over all 27 poses (Figure 3), inside the 0.002 Å acceptance tolerance of Section 4.5.

All three complexes have a pose below 2 Å (Table 3); for two of them it is the top-ranked pose. For 1HCK the best match is mode 2 (1.74 Å) while the top-ranked mode is only "acceptable" (2.74 Å), which is why both values are reported. In every complex the nine modes span a wide range (up to 12.2 Å for 1UWH), so a minimum over modes is much more favourable than a typical pose. Three complexes are a software-integration check; they do not measure Vina's pose-recovery rate.

Why no superposition. A rigid translation of a ligand leaves its shape unchanged, yet it moves it out of the binding site. Translating ethanol by 0–4 Å and evaluating RDKit's in-place `CalcRMS` returns an RMSD

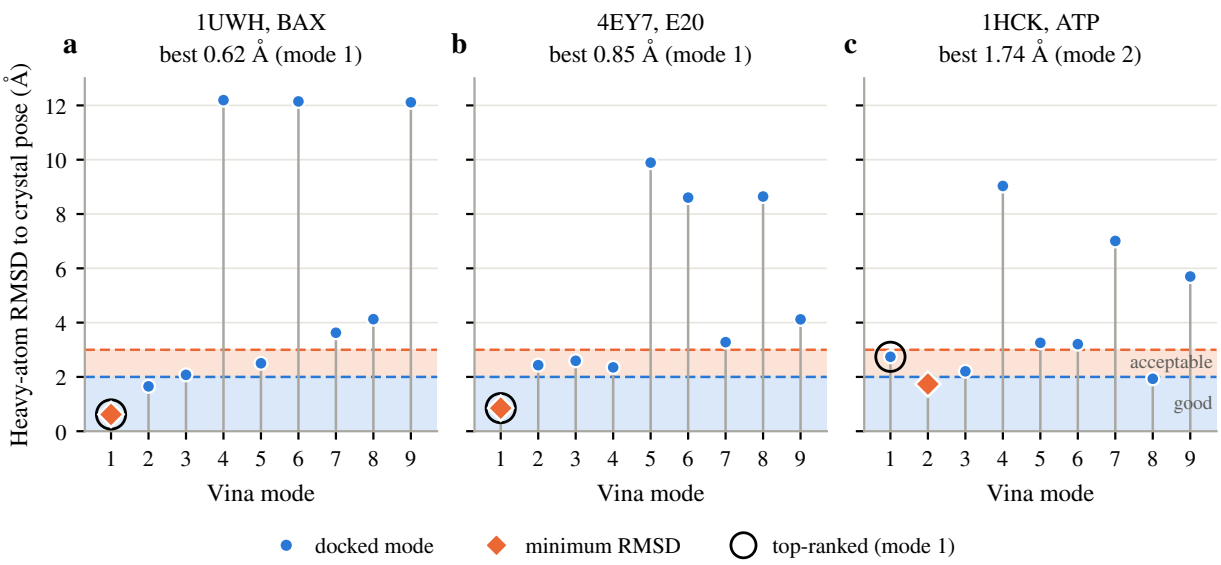


Figure 2. In-place heavy-atom RMSD of each Vina mode to the crystal pose for three native ligands. The orange diamond marks the pose with minimum RMSD and the ring marks the top-ranked pose (mode 1); the blue band is the "good" class (< 2 Å) and the orange band the "acceptable" class (2–3 Å) of Eq. (7). Data: 27 stored poses.

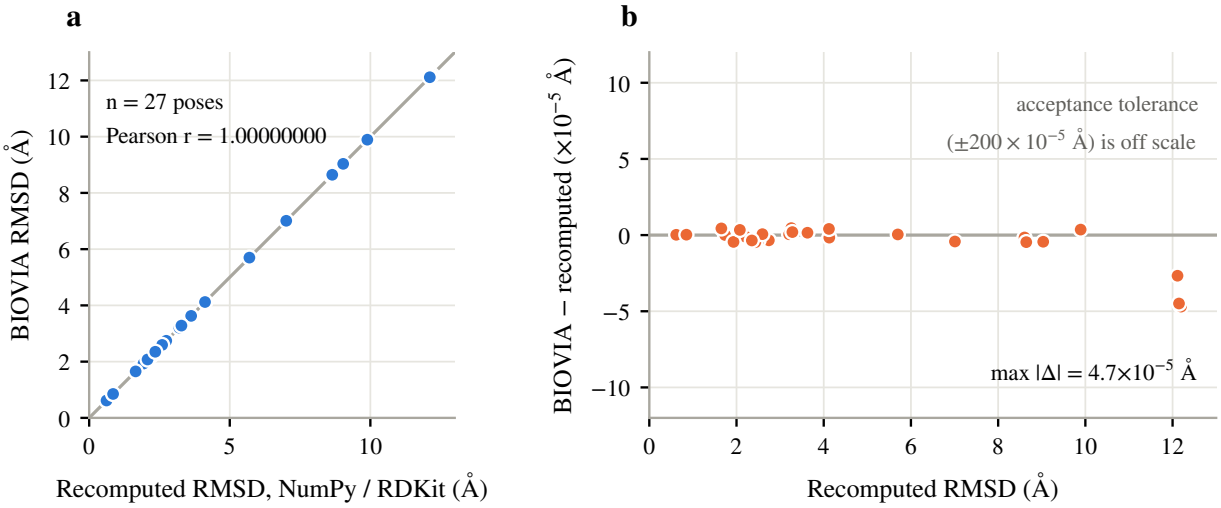


Figure 3. Agreement between the BIOVIA Discovery Studio RMSD and the recomputed in-place RMSD for the 27 poses. (a) Identity plot. (b) Residual (BIOVIA minus recomputed) in units of 10^{-5} Å; the largest magnitude is 4.7×10^{-5} Å, far inside the 0.002 Å tolerance.

equal to the shift (maximum deviation 2×10^{-16} Å), so the classes of Eq. (7) flip exactly at 2.0 and 3.0 Å (Figure 4). A superposition-based RMSD would return zero for every shift.

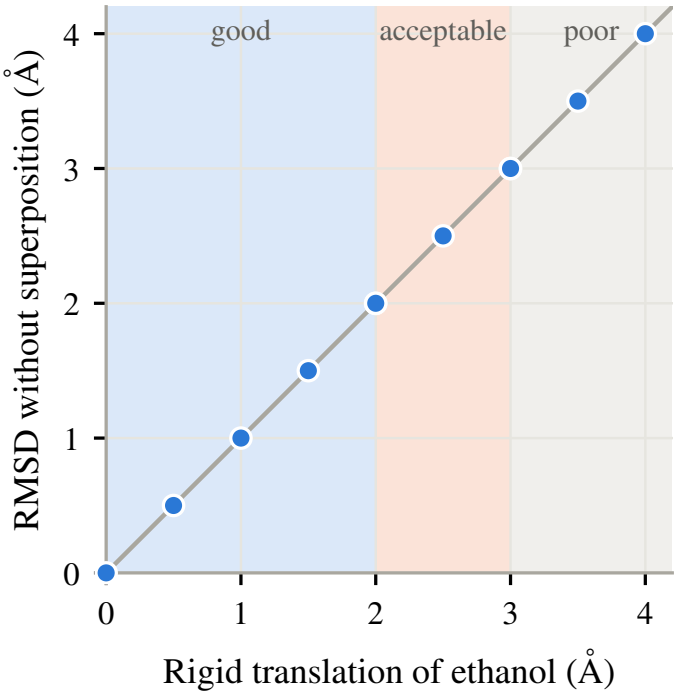


Figure 4. In-place RMSD of ethanol rigidly translated along one axis. The points fall on the identity line, and the class boundaries sit at 2.0 and 3.0 Å (shifts of exactly 2.0 Å are "acceptable"; 1.999 Å is "good").

5.3 Constructed demonstration: interaction similarity

Figure 5 works through Eqs. (8)–(11) for a constructed example: the native ligand contacts four residues with four (residue, type) pairs; the test ligand contacts four residues of which three coincide with the native set, but only two (residue, type) pairs coincide because it contacts Phe80 through a different interaction. Then $S_{\text{res}} = 3/4 = 75.0\%$, $S_{\text{type}} = 2/4 = 50.0\%$, and $S = 62.5\%$; with affinities -8.4 and $-7.9\,\text{kcal mol}^{-1}$, $\Delta G = -0.5\,\text{kcal mol}^{-1}$. The residue and type scores answer different questions and can diverge sharply, which is why both are reported.

The directional property can be seen by adding two spurious contacts to the test ligand: S is unchanged at 62.5%. If the test ligand contains every native contact plus two more, then $S(T, R) = 100\%$ but the reversed score $S(R, T) = 66.7\%$ (computed by swapping roles), so the score must not be read as a symmetric distance. The contacts here are invented for illustration; they are not measured interactions of a real complex.

5.4 Constructed demonstration: descriptor space

To show the descriptor/PCA stage, eight well-known compounds in two arbitrary panels (A: aspirin, acetaminophen, ibuprofen, naproxen; B: caffeine, quercetin, limonene, linalool) were described by molecular weight, logP, HBD and HBA using the package's own `LipinskiCalculator`; all eight have $V = 0$ in Eq. (3). Auto-scaled PCA (Eq. 13, $p = 1$) gives explained variances of 61.8%, 31.7% and 6.4% for PC1–PC3 (Figure 6), reproduced by an independent singular-value decomposition. Panel A compounds lie close together near the origin while panel B spreads widely, with quercetin (many donors and acceptors) at one extreme. The grouping is a property of these four descriptors and eight molecules; no 95% ellipses are drawn here because four points per group are too few for the covariance estimate to mean much.

5.5 Simulated demonstration: GC–MS module

Twelve total-ion chromatograms were *simulated* (fixed seed): 11 Gaussian peaks (standard deviation 0.045 min) on a drifting baseline with white noise and random retention-time shifts of ±0.12 min, for

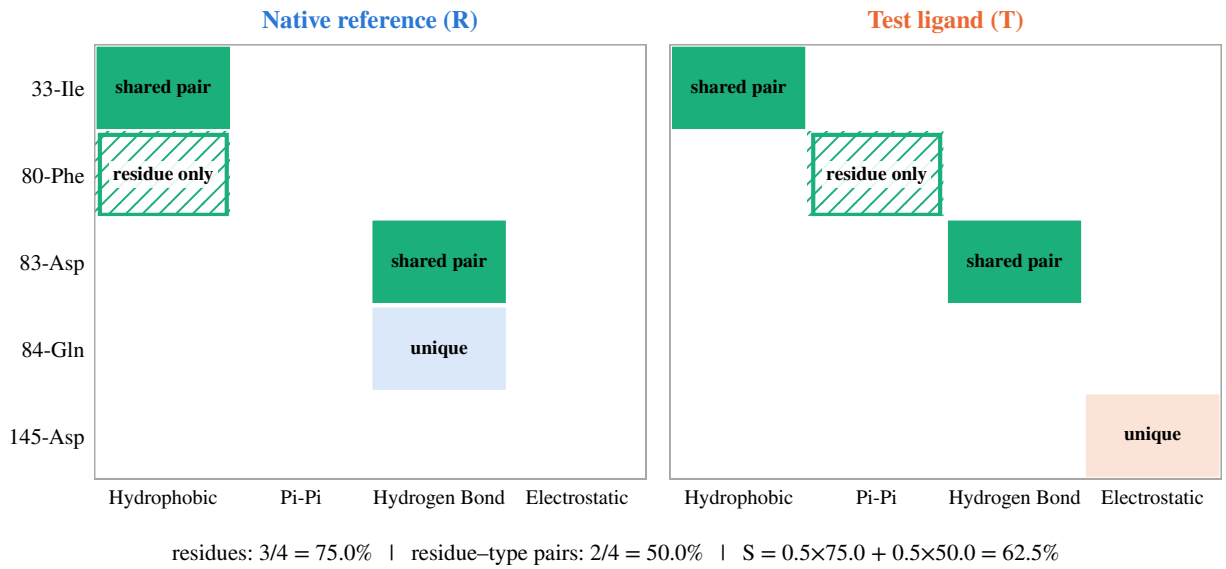


Figure 5. Worked example of the interaction-similarity score (constructed contacts). Green cells are (residue, type) pairs shared by both ligands, hatched cells are residues shared with a different interaction type, and tinted cells are unique to one ligand (blue: native; orange: test). The residue score counts residues in common (3 of 4) and the type score counts pairs in common (2 of 4).

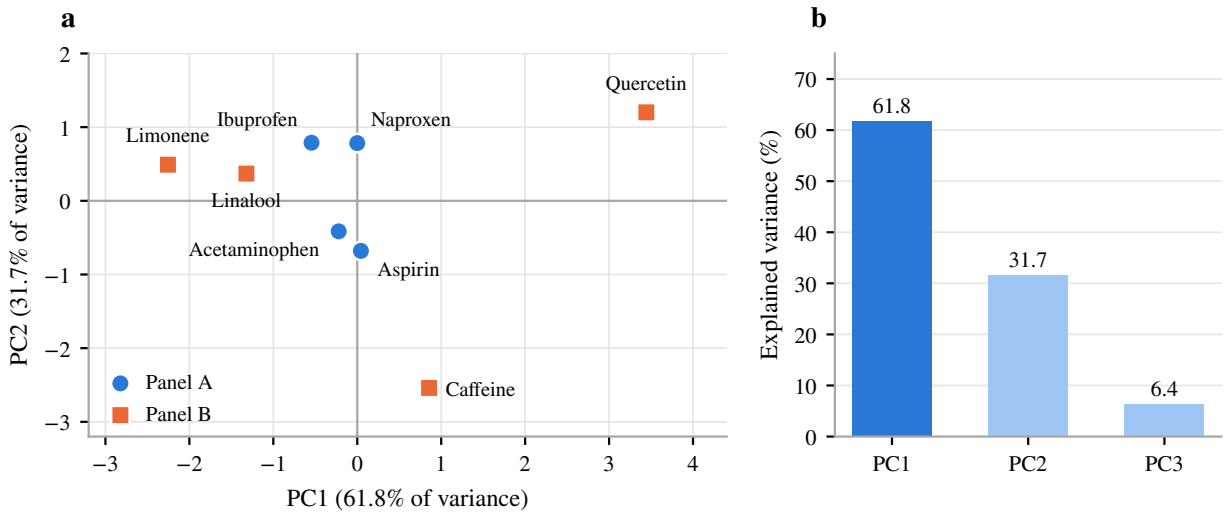


Figure 6. Descriptor PCA of eight reference compounds (illustration only). (a) Scores on PC1 and PC2. (b) Explained variance of the first three components (Eq. 14).

two classes (“Authentic” and “Adulterated”) and two series, three replicates each. Three analytes were deliberately changed in the Adulterated class: the peaks at 12.0 and 16.0 min were enriched by factors of 4.0 and 3.0, and the peak at 11.3 min was depleted to 0.3 of its height. The data were passed through `chemflow gcms` with total-sum normalisation, Pareto scaling, 199 permutations and seed 0 (smoothing window 0.05 min, minimum prominence 3% and presence in at least half the samples, chosen to suppress noise peaks).

The module recovered 11 aligned features (Figure 7a). PCA separates the classes along PC1 (71.3% of variance) and the two series along PC2 (24.8%) (Figure 7b). PLS-DA with two components achieved 100% five-fold cross-validated accuracy, $Q^2 = 0.993$ and permutation $p = 0.01$ (199 permutations); LDA also achieved 100% with $p = 0.01$. The three largest VIP scores belong to exactly the three altered peaks (12.0 min, 1.95; 16.0 min, 1.46; 11.3 min, 1.34; Figure 7c).

Two cautions follow from the simulation itself. First, perfect classification is built into the simulation

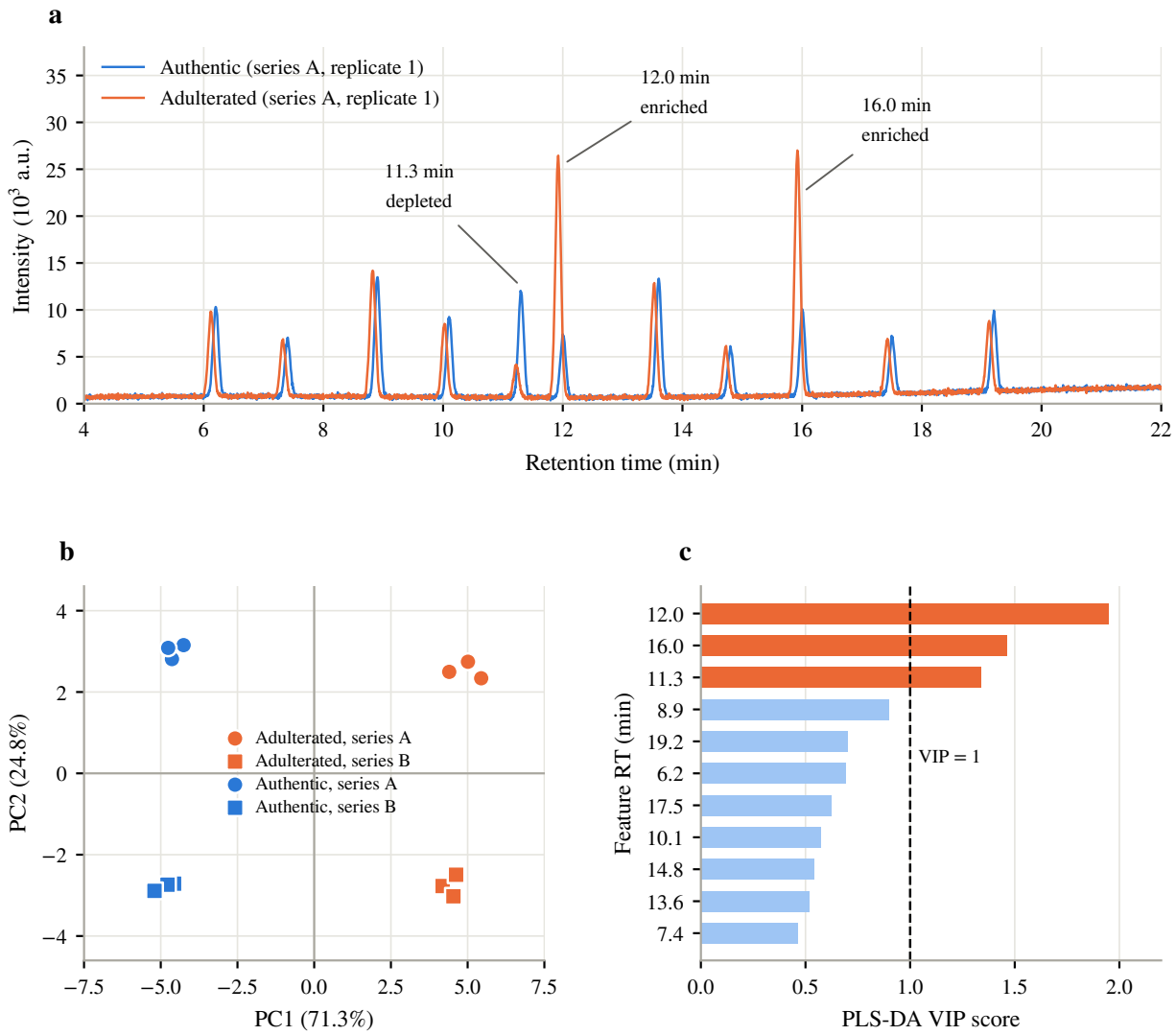


Figure 7. Simulated GC–MS demonstration. (a) Representative simulated chromatograms (series A, replicate 1) with the three deliberately changed peaks annotated. (b) PCA scores; colour is class (blue: Authentic; orange: Adulterated) and marker shape is series. (c) PLS-DA VIP scores; orange bars mark the three altered peaks, and the dashed line is VIP = 1.

design and says nothing about performance on real samples. Second, eight of the eleven features pass Welch $q < 0.05$ although only three were changed. The other five are a closure effect: total-sum normalisation forces each chromatogram to sum to 100, so enriching two peaks lowers the relative height of all the others, and these appear as differences. This is inherent to compositional normalisation, and it is why the VIP ranking, which concentrates on the three altered peaks, is more informative than the list of significant features. The supervised models also do not nest model selection inside cross-validation, and preprocessing is not repeated within folds (Section 7).

6 Discussion

Design outcomes. ChemFlow records provenance and makes conventions explicit. The redocking stage is part of every run, its in-place RMSD is cross-checked by an independent implementation, replicate seeds are deterministic, and fallbacks (force field, atom mapping, RMSD backend) are written to the output rather than hidden. The tests exercise the formulas of Section 4 directly, so a change that alters a definition is likely to be noticed.

Recommended practice.

- Report the RMSD of the best-affinity pose together with the minimum over poses (Table 3); for 1HCK they differ by 1.0 Å and cross a class boundary.
- State the standard-deviation convention. Eq. (5) uses R, so values are not interchangeable with spreadsheet defaults that use $R - 1$.
- Read the interaction-similarity score as "how much of the native contact pattern is reproduced", not as a distance, and report both S_{res} and S_{type}.
- When normalising GC–MS data by total sum, interpret univariate significance together with VIP ranks and, where possible, an absolute-abundance or internal-standard check.

Scope of the evidence. Docking accuracy, enrichment and hit rates have not been evaluated; doing so requires independent benchmark sets and prospective experiments.

7 Limitations

- **Chemistry.** One conformer per ligand; no protonation, tautomer or stereoisomer enumeration; largest-fragment selection and neutralisation can change the species docked; the PDB intermediate drops bond orders; minimiser convergence is not checked.
- **Receptor model.** The Kollman-style table is representative, not a complete force-field implementation; unlisted atoms default to zero charge; atom typing uses simple local rules; Vina does not use Coulomb electrostatics, so charge errors affect compatibility more than scores.
- **Validation scope.** Three complexes, one replicate each. A minimum over many poses is optimistic. The greedy atom-mapping fallback ignores topology and is approximate.
- **Similarity.** Directional; residue keys omit the chain; interaction tables are exported manually from a proprietary program, so this stage is Windows-only and needs an active desktop session. BIOVIA results come from archived runs and were not regenerated.
- **Statistics.** Replicate SD uses R in the denominator and is zero for one replicate. GC–MS supervised results do not use nested model selection and do not recompute preprocessing within cross-validation folds; compositional closure can create apparent differences.
- **Infrastructure.** ADMET retrieval depends on a third-party web service whose layout may change; the command-line scraper disables TLS certificate verification by default whereas the web interface enforces it; per-row ADMET mapping is positional when row counts match; checkpoints are keyed on file existence, not content hashes; automatic updates can change the version between sessions (they can be disabled).
- **Evidence.** Line coverage is not measured; demonstrations in Sections 5.3–5.5 use constructed or simulated data.

8 Availability

Software. ChemFlow 3.1.0 is released under the MIT licence; the source is at `https://github.com/Arifmaulanaazis/chemflow`. It requires Python $\geq$ 3.10 and, for ligand conversion, Open Babel; it installs with `pip` in editable mode (add the `web` extra for the web interface). The BIOVIA interaction stage additionally needs a licensed BIOVIA Discovery Studio installation on Windows.

Tools. The stored-pose RMSD re-evaluation of Section 5.2 is provided by `scripts/verify_rmsd_results.py`. GC–MS and chemometric analyses run through `chemflow gcms` and the `analytics/` modules.

Declarations

Author contributions. A.M.A. designed and implemented the software and wrote the manuscript. **Funding.** None declared. **Competing interests.** None declared. **Ethics.** No human or animal subjects were involved.

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